

A Six-Parameter Extension of the Direct Somos–4 Algebraic Generating Function

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Abstract

In an earlier paper we gave a five-parameter quadratic algebraic generating function whose Hankel determinants begin with (λ, m, r, η) and satisfy a normalized Somos–4 recurrence. Here we allow the coefficient of $H_{k+3}H_{k+1}$ to vary as well:

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2.$$

The main construction is simple. Introduce q with $q^6 = \alpha$, rescale the five old parameters by suitable powers of q , and replace x by qx . This produces the desired six-parameter Hankel sequence while leaving the prescribed first four determinants unchanged.

There is, however, an important distinction between the determinant sequence and the particular generating function used to produce it. The determinant sequence is rational in the six parameters, and so is a quartic model of the associated algebraic curve. The transported generating function is not: its coefficient of x is already $q^{-5}r$. Thus the determinants and the curve can be written without adjoining a root of α , even though this particular power-series representation cannot.

We compare this construction with the general quadratic-orthogonal-pair construction of Chang and Liu. After a direct shift, their bilateral Somos–4 sequence agrees with ours, and their subdiagonal parameters satisfy $d_n = B_{n+2}$. Their square-root choice $u^2 = \alpha$ also has a concrete meaning. Replacing u by $-u$ is exactly the same as replacing a generating series $F(x)$ by $F(-x)$. This leaves every ordinary Hankel determinant unchanged, but multiplies the once-shifted determinant of order N by $(-1)^N$. In this sense the ordinary Hankel sequence forgets a parity choice that the shifted determinants still remember.

1 Introduction

The preceding paper [3] starts with four prescribed Hankel determinants,

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta),$$

and constructs an algebraic generating function for which

$$H_{k+4}H_k = H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{1}$$

Here we ask what happens when the coefficient of the first product is no longer fixed to 1:

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{2}$$

We therefore have six independent parameters $(\lambda, m, r, \eta, \alpha, \tau)$. Setting $\alpha = 1$ takes us back to the construction of the preceding paper.

There are two questions, and it is useful not to mix them. The first is an existence question: can we extend the old construction so that the first four Hankel determinants remain exactly (λ, m, r, η) and the recurrence is (2)? The answer is yes. Section 2 gives an explicit weighted rescaling.

The second question asks for more. Can such a quadratic algebraic generating function be written using only rational functions of the six parameters, that is, over

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau),$$

while also keeping the shifted-determinant identity $D_N = H_{N+2}$? Our weighted construction does not do this. It uses a sixth root q of α , and its coefficient of x is $q^{-5}r$. This is not a cosmetic radical that disappears after simplification.

A small vocabulary point will make the rest of the paper easier to read. By an *ordinary Hankel sequence* we mean the determinants H_N of a coefficient sequence. A *once-shifted determinant* D_N uses the same coefficients with every index increased by one. We sometimes call a chosen coefficient sequence a *moment sequence*, following the standard language of continued fractions and orthogonal polynomials. No analytic measure is needed here: throughout this paper, “moments” are simply the coefficients of the formal generating series. Likewise, the *Somos-4 orbit* means the sequence obtained by iterating the recurrence once its initial values are fixed.

One theme of the paper is that the following three objects carry different amounts of information:

Hankel determinant sequence, algebraic curve, chosen generating series.

The first two can be written over K . A particular generating series need not be. We will use *descend to K* as shorthand for “can be written with coefficients in K ”. With this terminology, the determinant sequence and the curve descend, while the particular transported series does not.

This distinction also explains our relation to recent work. Liu, Wang, and Zhang [2] give broad rational-function-field constructions for general (α, β) -Somos-4 Hankel transforms. Here we impose the additional requirements that four initial determinants are independently prescribed and that a specific shifted-determinant normalization is retained. So our question is narrower.

The paper is organized as follows. Section 2 performs the weighted rescaling and proves the six-parameter recurrence. Section 3 tracks the shifted determinants and shows exactly where the fractional power of α first appears. Section 4 shows that the determinant sequence and its curve can still be written over K . Section 5 translates the construction into the notation of Chang and Liu and identifies the corresponding Jacobi subdiagonal data. Section 6 explains the remaining sign choice as the elementary parity operation $F(x) \mapsto F(-x)$. Section 7 collects the main conclusions.

2 From the five-parameter construction to six parameters

2.1 The five-parameter input

Let

$$G_{1;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} g_k x^k$$

be the selected algebraic series of [3]. Define

$$H_N(g) = \det(g_{i+j})_{0 \leq i,j \leq N-1}, \quad D_N(g) = \det(g_{i+j+1})_{0 \leq i,j \leq N-1}.$$

The normalization in the preceding paper is

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta), \quad D_N = H_{N+2}. \tag{3}$$

Its quadratic discriminant factors as

$$\Delta_1(x) = (x\lambda^2\eta mr^2)^2 \left((1 - Jx + \tau x^2)^2 - 4x^3 \right), \tag{4}$$

where

$$J = \frac{\lambda\eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda\eta}. \quad (5)$$

We will use only these two facts from the five-parameter construction.

2.2 Weighted transport

Work over

$$\mathbb{K} = \mathbb{Q}(\lambda, m, r, \eta, q, \tau), \quad q^6 = \alpha,$$

and introduce the primed parameters

$$\lambda' = \lambda, \quad m' = q^{-2}m, \quad r' = q^{-6}r, \quad \eta' = q^{-12}\eta, \quad \tau' = q^{-8}\tau. \quad (6)$$

Define

$$G_{\alpha;\lambda,m,r,\eta,\tau}(x) := G_{1;\lambda',m',r',\eta',\tau'}(qx). \quad (7)$$

If

$$G_{1;\lambda',m',r',\eta',\tau'}(x) = \sum_{k \geq 0} a_k x^k, \quad G_{\alpha;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} b_k x^k,$$

then

$$b_k = q^k a_k. \quad (8)$$

Proposition 1 (Determinant weights). *For every $N \geq 1$,*

$$H_N(b) = q^{N(N-1)} H_N(a), \quad D_N(b) = q^{N^2} D_N(a). \quad (9)$$

Proof. In the ordinary Hankel matrix, its (i, j) entry is multiplied by q^{i+j} . Factoring q^i from row i and q^j from column j gives the exponent $2 \sum_{j=0}^{N-1} j = N(N-1)$. For the shifted matrix the entry is multiplied by q^{i+j+1} . The additional factor q in every row contributes N , so the total exponent is $N(N-1) + N = N^2$. $\square$

2.3 The six-parameter Somos-4 recurrence

Theorem 2. *Let $\hat{H}_N = H_N(b)$. Then*

$$(\hat{H}_1, \hat{H}_2, \hat{H}_3, \hat{H}_4) = (\lambda, m, r, \eta)$$

and

$$\hat{H}_{k+4} \hat{H}_k = \alpha \hat{H}_{k+3} \hat{H}_{k+1} + \tau \hat{H}_{k+2}^2, \quad k \geq 1. \quad (10)$$

Proof. Equation (9) and (6) give

$$\hat{H}_1 = \lambda, \quad \hat{H}_2 = q^2 m' = m, \quad \hat{H}_3 = q^6 r' = r, \quad \hat{H}_4 = q^{12} \eta' = \eta.$$

Write $H'_N = H_N(a)$. The primed sequence satisfies

$$H'_{k+4} H'_k = H'_{k+3} H'_{k+1} + \tau' H_{k+2}'^2.$$

The weight difference between the left-hand product and the first product on the right is 6, while that between the left-hand product and $H_{k+2}'^2$ is 8. Hence

$$\hat{H}_{k+4} \hat{H}_k = q^6 \hat{H}_{k+3} \hat{H}_{k+1} + q^8 \tau' \hat{H}_{k+2}^2.$$

Now use $q^6 = \alpha$ and $q^8 \tau' = \tau$. $\square$

Corollary 3 (Low-order recurrence check). *The first two values forced beyond the initial tuple are*

$$\hat{H}_5 = \frac{\alpha \eta m + \tau r^2}{\lambda}, \quad \hat{H}_6 = \frac{\alpha \hat{H}_5 r + \tau \eta^2}{m}.$$

These formulas provide a direct indexing check for any coefficient expansion of (7).

3 Why fractional powers appear

The rescaling above gives exactly the ordinary Hankel determinants we want. The shifted determinants scale differently, however. We compute that difference next; it is the source of the stronger rationality question.

3.1 The transported companion identity

Proposition 4. *If the primed series obeys $D_N(a) = H'_{N+2}$, then*

$$\widehat{D}_N := D_N(b) = q^{-3N-2} \widehat{H}_{N+2}, \quad N \geq 1. \quad (11)$$

Proof. By (9),

$$D_N(b) = q^{N^2} H'_{N+2}, \quad \widehat{H}_{N+2} = q^{(N+2)(N+1)} H'_{N+2}.$$

Their exponent difference is $N^2 - (N+2)(N+1) = -3N-2$. $\square$

Remark 5. A constant coefficient rescaling $b_k \mapsto cb_k$ changes the ratio $D_N(b)/\widehat{H}_{N+2}$ by c^{-2} . It cannot remove the factor q^{-3N} . Thus the literal identity $D_N = H_{N+2}$ is not invariant under this graded transport. Equation (11), rather than the ungraded identity, is the companion normalization naturally inherited from the five-parameter series.

3.2 Parity conjugation and the companion orientation

For any formal series

$$F(x) = \sum_{n \geq 0} f_n x^n, \quad P(F)(x) := F(-x) = \sum_{n \geq 0} (-1)^n f_n x^n,$$

the involution P has different effects on the ordinary and once-shifted Hankel determinants.

Proposition 6 (Parity determinant laws). *For every $N \geq 1$,*

$$H_N(P(F)) = H_N(F), \quad D_N(P(F)) = (-1)^N D_N(F). \quad (12)$$

Proof. The (i, j) entry of the ordinary Hankel matrix is multiplied by $(-1)^{i+j}$. Factoring $(-1)^i$ from row i and $(-1)^j$ from column j gives the total exponent $2 \sum_{j=0}^{N-1} j = N(N-1)$, which is even. In the once-shifted matrix the entry is multiplied by $(-1)^{i+j+1}$. The additional factor -1 in every row contributes $(-1)^N$, proving the second identity. $\square$

Corollary 7 (Parity-conjugate companion law). *If $D_N(F) = c_N H_{N+2}(F)$, then*

$$D_N(P(F)) = (-1)^N c_N H_{N+2}(P(F)). \quad (13)$$

In particular, the ordinary Hankel orbit cannot distinguish $F(x)$ from $F(-x)$, whereas a prescribed companion law distinguishes the two parity orientations whenever some odd-order companion determinant is nonzero.

Applied to the transported series, parity conjugation preserves the six-parameter Somos-4 orbit and changes the graded companion identity to

$$D_N(G_\alpha(-x)) = (-1)^N q^{-3N-2} H_{N+2}(G_\alpha(-x)). \quad (14)$$

So the effect is quite concrete: $F(x)$ and $F(-x)$ have the same ordinary Hankel determinants, but their shifted determinants differ by a predictable sign. We use the word “orientation only for this sign choice. Nothing here requires an interpretation in terms of algebraic sheets or reversal of a bilateral Somos orbit.

3.3 The first coefficients and failure of descent

For the five-parameter selected series, the companion normalization gives $g_0 = \lambda$ and $g_1 = r$. Applied to the primed parameters in (6), this yields $a_0 = \lambda$ and $a_1 = r' = q^{-6}r$. Therefore (8) gives

$$b_0 = \lambda, \quad b_1 = q^{-5}r = \frac{qr}{\alpha}. \quad (15)$$

The next two coefficients obtained by direct expansion are

$$b_2 = \frac{m}{\lambda} + \frac{q^{-10}r^2}{\lambda} = \frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda}, \quad (16)$$

$$b_3 = \frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2}. \quad (17)$$

Hence

$$\begin{aligned} G_\alpha(x) = & \lambda + \frac{qr}{\alpha}x + \left(\frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda} \right) x^2 \\ & + \left(\frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2} \right) x^3 + O(x^4). \end{aligned} \quad (18)$$

The first coefficient also follows from (11) at $N = 1$:

$$b_1 = D_1(b) = q^{-5}H_3(b) = q^{-5}r.$$

Thus the fractional power is not an artifact of a badly simplified formula. It is already forced by the shifted-determinant relation at $N = 1$.

Put

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau), \quad L = K(q), \quad q^6 = \alpha.$$

Proposition 8 (Non-descent of the transported branch). *Regard α as an indeterminate. In the generic nondegenerate case $r \neq 0$, the transported series (7) does not belong to $K[[x]]$.*

Proof. Its coefficient of x is $b_1 = q^{-5}r = qr/\alpha$. The polynomial $X^6 - \alpha$ is irreducible over K by Eisenstein's criterion at the prime α in the corresponding polynomial ring. Hence $q \notin K$. Since $r \neq 0$ generically, $b_1 \notin K$, and the transported series is not in $K[[x]]$. $\square$

Although the coefficients do not descend, the fractional powers cancel in the ordinary Hankel determinants. For example,

$$H_2(b) = b_0b_2 - b_1^2 = \lambda \left(\frac{m}{\lambda} + \frac{q^{-10}r^2}{\lambda} \right) - q^{-10}r^2 = m.$$

This example shows the distinction directly. The radicals cancel from the ordinary Hankel determinants even though they remain in the coefficients of the transported series.

4 What is defined over the six-parameter field

The failure just proved concerns one particular generating series. It does not mean that the recurrence itself, or its algebraic curve, needs fractional powers of α . We now check these two objects separately.

4.1 Transport of the discriminant

Separate the old invariant into

$$A = \frac{\lambda\eta}{mr} + \frac{\tau mr}{\lambda\eta}, \quad B = \frac{m^2}{\lambda r} + \frac{r^2}{m\eta}. \quad (19)$$

After substitution of (6),

$$J' = q^{-4}A + q^2B. \quad (20)$$

Consequently the substitution $x \mapsto qx$ gives

$$qJ' = q^{-3}A + q^3B = \alpha^{-1/2}A + \alpha^{1/2}B. \quad (21)$$

Proposition 9 (Transported discriminant). *The discriminant obtained by substituting (6) and $x \mapsto qx$ into (4) is*

$$\Delta_\alpha(x) = q^{-50}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (q^{-3}A + q^3B)x + q^{-6}\tau x^2 \right)^2 - 4q^3x^3 \right]. \quad (22)$$

Equivalently,

$$\Delta_\alpha(x) = \alpha^{-25/3}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (\alpha^{-1/2}A + \alpha^{1/2}B)x + \frac{\tau}{\alpha}x^2 \right)^2 - 4\alpha^{1/2}x^3 \right]. \quad (23)$$

Proof. The square prefactor has total q -weight $2(1 - 12 - 2 - 12) = -50$. The three terms in the residual factor transform as $J'x \mapsto qJ'x$, $\tau'x^2 \mapsto q^2\tau'x^2 = q^{-6}\tau x^2$, and $x^3 \mapsto q^3x^3$. Equations (20) and $q^6 = \alpha$ give the stated forms. $\square$

The overall monomial factor in (22) is a square in $\mathbb{K}(x)$ and does not affect the residual double cover. It should, however, be retained when comparing discriminants of particular polynomial representatives, because multiplying a quadratic equation by a scalar squares that scalar in its discriminant.

4.2 A rational model of the curve

Remove the square prefactor in (23) and put

$$x = \alpha^{1/2}z. \quad (24)$$

The residual equation becomes

$$Y^2 = \left(1 - (A + \alpha B)z + \tau z^2 \right)^2 - 4\alpha^2 z^3. \quad (25)$$

All coefficients of (25) belong to $\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$.

For comparison with the quartic convention used in [3], set $X = z^{-1}$ and $y = z^{-2}Y$ (equivalently, $Y = z^2y$). Then (25) is birational to

$$y^2 = \left(X^2 - (A + \alpha B)X + \tau \right)^2 - 4\alpha^2 X. \quad (26)$$

Thus the curve itself has coefficients in the rational six-parameter field. But the change of variable (24) also rescales the coefficients of a power series, so it does not preserve our normalized Hankel data. Removing radicals from the curve is therefore not the same as removing them from the chosen generating series.

4.3 Formal representatives with the same Hankel determinants

The ordinary Hankel determinants do not determine a unique generating series. This is worth making explicit. For a nondegenerate Hankel sequence define

$$\beta_n = \frac{H_{n-1}H_{n+1}}{H_n^2}, \quad H_0 = 1. \quad (27)$$

All H_n generated by (10), and hence all β_n , lie in K . For arbitrary $a_0, a_1, \dots \in K$, the Jacobi fraction

$$\tilde{G}(x) = \frac{H_1}{1 - a_0x - \frac{\beta_1x^2}{1 - a_1x - \frac{\beta_2x^2}{1 - a_2x - \ddots}}} \quad (28)$$

has ordinary Hankel determinants H_n . So there are formal power series over K with exactly the same ordinary Hankel determinants. What we do *not* get for free is the extra structure we care about: such a series need not be quadratic algebraic, need not satisfy the graded shifted-determinant identity, and need not be the distinguished Stieltjes choice.

5 Comparison with the construction of Chang and Liu

We have shown only that the transported quadratic series contains fractional powers. A different quadratic series might conceivably avoid them. Chang and Liu provide a natural comparison because their construction realizes a generic bilateral Somos–4 sequence by Hankel determinants. We now put their formulas and ours into the same notation.

5.1 The shifted Somos–4 orbit

Chang and Liu [1, Sections 3.3–4.2] treat the generic bilateral Somos–4 initial-value problem by Hankel determinants of quadratic orthogonal pairs. Their normalized recurrence is

$$\sigma_{n+2}\sigma_{n-2} = a\sigma_{n+1}\sigma_{n-1} + b\sigma_n^2, \quad n \in \mathbb{Z}, \quad (29)$$

with $\sigma_{-1} = x$, $\sigma_0 = 1$, $\sigma_1 = y$, and $\sigma_2 = z$. Our freely prescribed determinants fit this normalization after the shift

$$\sigma_n = \frac{H_{n+2}}{m}, \quad (x, y, z; a, b) = \left(\frac{\lambda}{m}, \frac{r}{m}, \frac{\eta}{m}; \alpha, \tau \right). \quad (30)$$

This shift uses all six parameters. In particular, it avoids the unwanted constraint $\eta = \alpha\lambda r + \tau m^2$ that would follow from imposing our recurrence at $N = 0$ with $H_0 = 1$.

Write $u^2 = \alpha$ and define

$$\mathcal{N} = \tau r^2 m + \lambda^2 \eta^2 + \alpha \eta m^3 + \alpha \lambda r^3, \quad (31)$$

$$\mathcal{N}_0 = \tau r^2 m - \lambda^2 \eta^2 + \alpha \eta m^3 + \alpha \lambda r^3. \quad (32)$$

Under (30), the Chang–Liu parameters include

$$d_0 = \frac{\lambda r}{m^2}, \quad u_0 = \frac{2r}{m}, \quad (33)$$

$$\frac{v}{u} = \frac{\mathcal{N}}{2\alpha\lambda r\eta m}, \quad uv = \frac{\mathcal{N}}{2\lambda r\eta m}, \quad (34)$$

$$\frac{v_0}{u} = \frac{\mathcal{N}_0}{2\alpha\lambda r\eta m}, \quad uv_0 = \frac{\mathcal{N}_0}{2\lambda r\eta m}, \quad (35)$$

and

$$f = \frac{\tau}{\alpha} - \frac{\mathcal{N}^2}{4\alpha\lambda^2 r^2 \eta^2 m^2}. \quad (36)$$

The sign difference between $\mathcal{N}$ and $\mathcal{N}_0$ is essential because v_0 enters the distinguished moment branch. The scaling $X = uZ$ turns their curve into

$$Y^2 = (\alpha Z^2 + f)^2 + 4\alpha Z - 4uv, \quad (37)$$

which is defined over K . Again, this statement concerns the curve only. It says nothing yet about whether the coefficients of a chosen generating series lie in K .

5.2 Translation to Jacobi data

Use the local Jacobi convention

$$F(t) = \frac{\lambda}{1 - A_0 t - \frac{B_1 t^2}{1 - A_1 t - \frac{B_2 t^2}{1 - A_2 t - \frac{B_3 t^2}{1 - A_3 t - B_4 t^2 M_4(t)}}}}, \quad M_4(0) = 1. \quad (38)$$

The product formula

$$H_N = \lambda^N \prod_{j=1}^{N-1} B_j^{N-j} \quad (39)$$

gives

$$B_1 = \frac{m}{\lambda^2}, \quad B_2 = \frac{\lambda r}{m^2}, \quad B_3 = \frac{m\eta}{r^2}, \quad B_4 = \frac{r(\alpha\eta m + \tau r^2)}{\lambda\eta^2}. \quad (40)$$

The companion identities through $D_3 = H_5$ force

$$A_0 = \frac{r}{\lambda}, \quad A_1 = \frac{\eta\lambda^2 + m^2}{\lambda m r}, \quad A_2 = \frac{\alpha\eta m^3 + \lambda r^3 + \tau m^2 r^2}{\eta\lambda m r}. \quad (41)$$

Proposition 10 (Fourth shifted determinant). *For the convention (38),*

$$D_4 = \lambda^4 B_1^3 B_2^2 B_3 (A_0 A_1 A_2 A_3 - A_0 A_1 B_3 - A_0 A_3 B_2 - A_2 A_3 B_1 + B_1 B_3). \quad (42)$$

After substitution of (40) and (41), this reduces to

$$D_4 = \frac{\eta}{\lambda r^2} \left[A_3 r (\alpha\eta m + \tau r^2) - \eta^2 \lambda \right]. \quad (43)$$

Consequently $D_4 = H_6$, where

$$H_6 = \frac{\alpha^2 \eta m r + \alpha \tau r^3 + \tau \eta^2 \lambda}{\lambda m},$$

uniquely forces

$$A_3 = \frac{\alpha^2 \eta m r^3 + \alpha \tau r^5 + \eta^3 \lambda m + \tau \eta^2 \lambda r^2}{\eta m r (\alpha\eta m + \tau r^2)}. \quad (44)$$

Proof. Equation (42) follows by direct expansion of the shifted 4×4 Hankel determinant in Jacobi parameters. Substitution and collection of the terms linear in A_3 gives (43). Solving (43) for A_3 after inserting the displayed value of H_6 gives (44). $\square$

An equivalent useful form is

$$A_3 = \frac{\alpha r^2}{\eta m} + \frac{\eta \lambda (\eta m + \tau r^2)}{m r (\alpha \eta m + \tau r^2)}.$$

No B_4 occurs in D_4 : the fourth ordinary Hankel step determines B_4 , while the fourth shifted step determines A_3 , without imposing any extra relation among the six parameters.

Chang and Liu use

$$d_n = \frac{\sigma_{n-1} \sigma_{n+1}}{\sigma_n^2}.$$

Substitution of (30) gives the exact all-order interface

$$d_n = \frac{H_{n+1} H_{n+3}}{H_{n+2}^2} = B_{n+2}. \quad (45)$$

Thus $d_0 = B_2$, $d_1 = B_3$, and $d_2 = B_4$. Thus the subdiagonal parameters match after an index shift: only B_1 comes before the Chang–Liu sequence begins. The diagonal parameters and the normalization of the generating-series coefficients still have to be treated separately.

6 The parity involution in the Chang–Liu branch

The Chang–Liu construction contains the same elementary parity symmetry, although it appears there as a choice of square root. Let

$$S(t; u) = \sum_{n \geq 0} s_n(u) t^n, \quad u^2 = \alpha,$$

be their distinguished moment branch. Chang and Liu, Remark 4.7, give

$$S(t; -u) = S(-t; u), \quad s_n(-u) = (-1)^n s_n(u). \quad (46)$$

Thus the parameter involution $u \mapsto -u$ is exactly the moment-parity involution of Proposition 12. Consequently,

$$H_N(S(t; -u)) = H_N(S(t; u)), \quad D_N(S(t; -u)) = (-1)^N D_N(S(t; u)). \quad (47)$$

So the two choices of u do not produce two different ordinary Somos–4 sequences. They produce coefficient sequences related by alternating signs. Ordinary Hankel determinants cannot see this change; once-shifted determinants can.

We can now see the same fact directly in the coefficients. Writing

$$S(t; u) = E(t^2) + utR(t^2),$$

the relation (46) implies, coefficientwise, that the even moments are invariant under $u \mapsto -u$ and the odd moments are anti-invariant. The appearance of $u = \sqrt{\alpha}$ in this selected branch therefore records the parity orientation of its odd moments. This observation concerns the selected moment lift. It does not assert that every representation of the ordinary Hankel orbit requires a quadratic extension.

The shifted-determinant normalization tells the two choices apart. If one branch satisfies $D_N = c_N H_{N+2}$, then its parity conjugate satisfies $D_N = (-1)^N c_N H_{N+2}$. Hence choosing a literal or graded companion law selects one of the two parity orientations. This is all we need from the involution.

7 Summary

The main facts can now be stated compactly.

1. The weighted transport (7) produces a quadratic algebraic series over $K(q)$, $q^6 = \alpha$, whose first four ordinary Hankel determinants are (λ, m, r, η) and whose complete ordinary Hankel orbit satisfies (10).
2. Its once-shifted determinants satisfy the exact graded companion law (11). The coefficient $b_1 = q^{-5}r$ proves generic non-descent of this selected transported branch to $K[[x]]$.
3. The associated residual curve has the rational model (25) over K . Thus descent of the orbit and the curve does not imply descent of the selected moment branch.
4. The dictionary (30) identifies the normalized Chang–Liu orbit with the prescribed six-parameter orbit, and (45) gives the all-order subdiagonal interface $d_n = B_{n+2}$.
5. The involution $u \mapsto -u$ in the Chang–Liu moment branch is precisely moment parity. It fixes every ordinary Hankel determinant and multiplies the once-shifted determinant of order N by $(-1)^N$.

8 Conclusion

There is a simple construction and a simple warning.

The construction is the weighted substitution (7). With $q^6 = \alpha$, it extends the five-parameter generating function to the full six-parameter Somos–4 recurrence. Its first four ordinary Hankel determinants remain (λ, m, r, η) , and the whole determinant sequence satisfies (10).

The warning is that the coefficients of the generating series contain more information than its ordinary Hankel determinants. Under the same rescaling, the shifted determinants obey (11), and already $b_1 = q^{-5}r$ lies outside $K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$ in the generic case. Nevertheless the ordinary Hankel sequence lies in K , and the quartic (25) shows that the associated curve does too. Thus three superficially similar rationality questions have different answers: the determinant sequence and the curve are rational in the six parameters, while this particular generating series is not.

The comparison with Chang and Liu makes the remaining square-root choice easy to interpret. Their identity $S(t; -u) = S(-t; u)$ says that changing the sign of $u = \sqrt{\alpha}$ merely alternates the signs of the odd coefficients. This leaves every ordinary Hankel determinant fixed and changes a once-shifted determinant of order N by $(-1)^N$. In other words, ordinary Hankel determinants forget this parity choice, while shifted determinants remember it. No stronger geometric interpretation is required for the results proved here.

Disclosure on authoring and verification tools

The manuscript was revised with assistance from AI-based language and symbolic calculation tools. Their use was directed toward notation control, LaTeX editing, and exact algebraic checks. The author remains responsible for the mathematical statements and the final text.

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